



The Fetal Membranes and Placenta

Learning Objectives



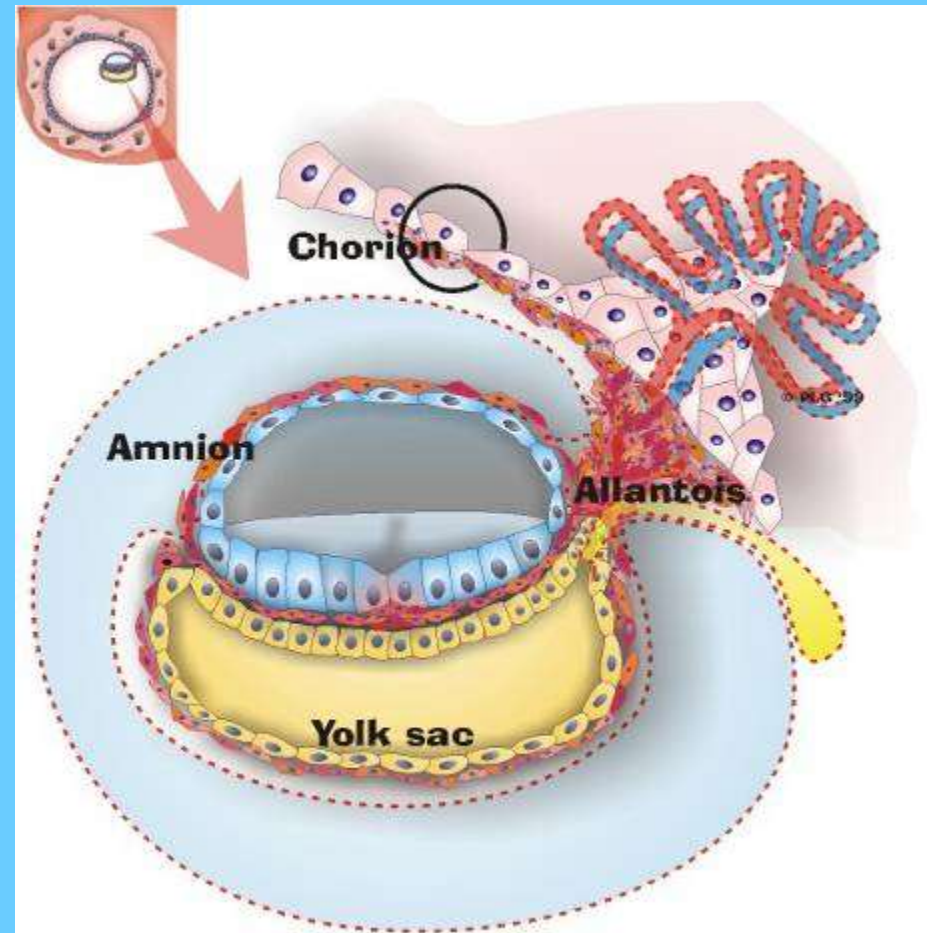
After completion of this lecture, the student should be able to:

- Understand the development and function of fetal membranes including chorion, amnion, yolk sac , umbilical cord and allantois.
- Discuss the birth injuries related to amniotic fluid.
- Give the development of placenta.
- understand how the placenta functions to maintain the fetus.



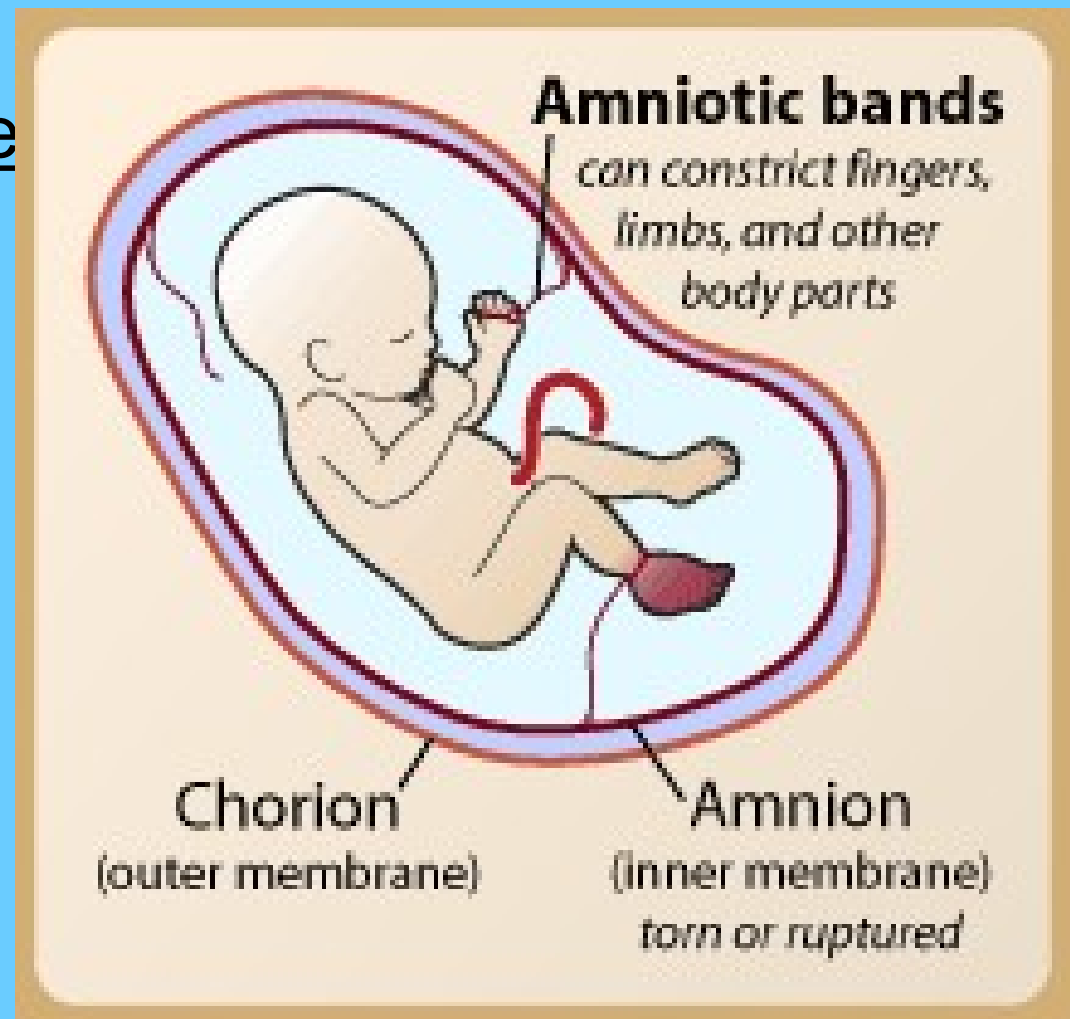
Fetal Membrane

- Any membrane that functions for the protection or nourishment or respiration or excretion of a developing fetus is known as a fetal membrane.
- Chorion ,
- Amnion ,
- Umbilical Vesicle and
- Allantois.



Chorion

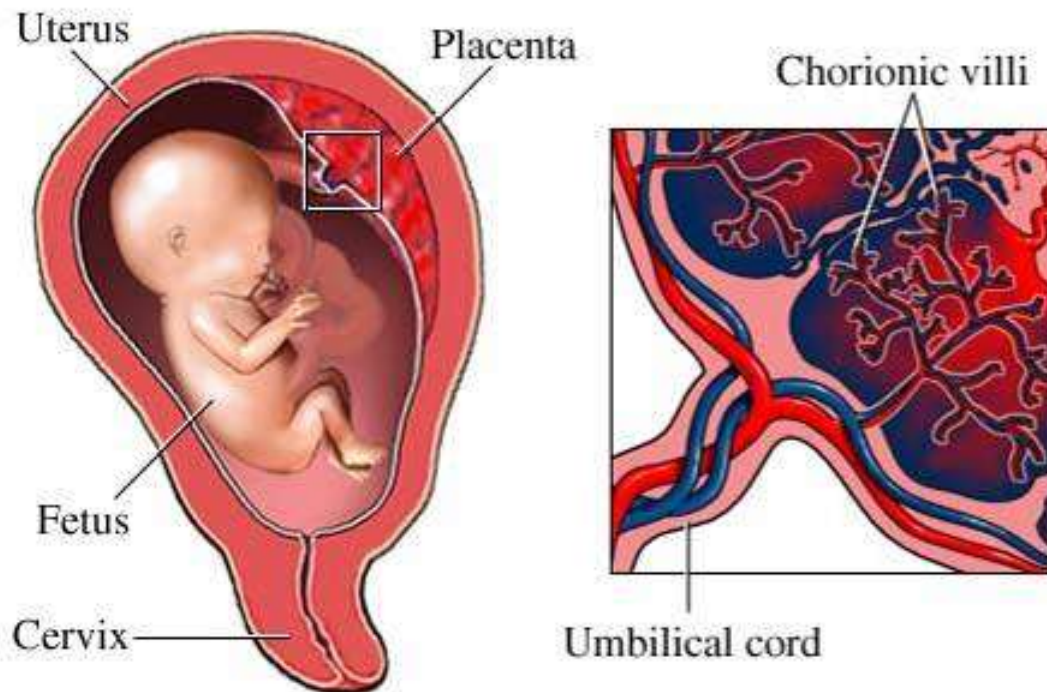
- One of the membranes that exist during pregnancy between the developing **fetus** and **mother**.
- Formed by :
 - Extraembryonic mesoderm and
 - Two layers of trophoblast.
- Surrounds the **embryo** and other membranes.





Chorionic Villi

- Chorion forms fingerlike projections called **villi**.
- **Chorion frondosum** where the villi are present (fetal part of the placenta).
- where the villi disappear and is smooth is called **chorion laeve**.

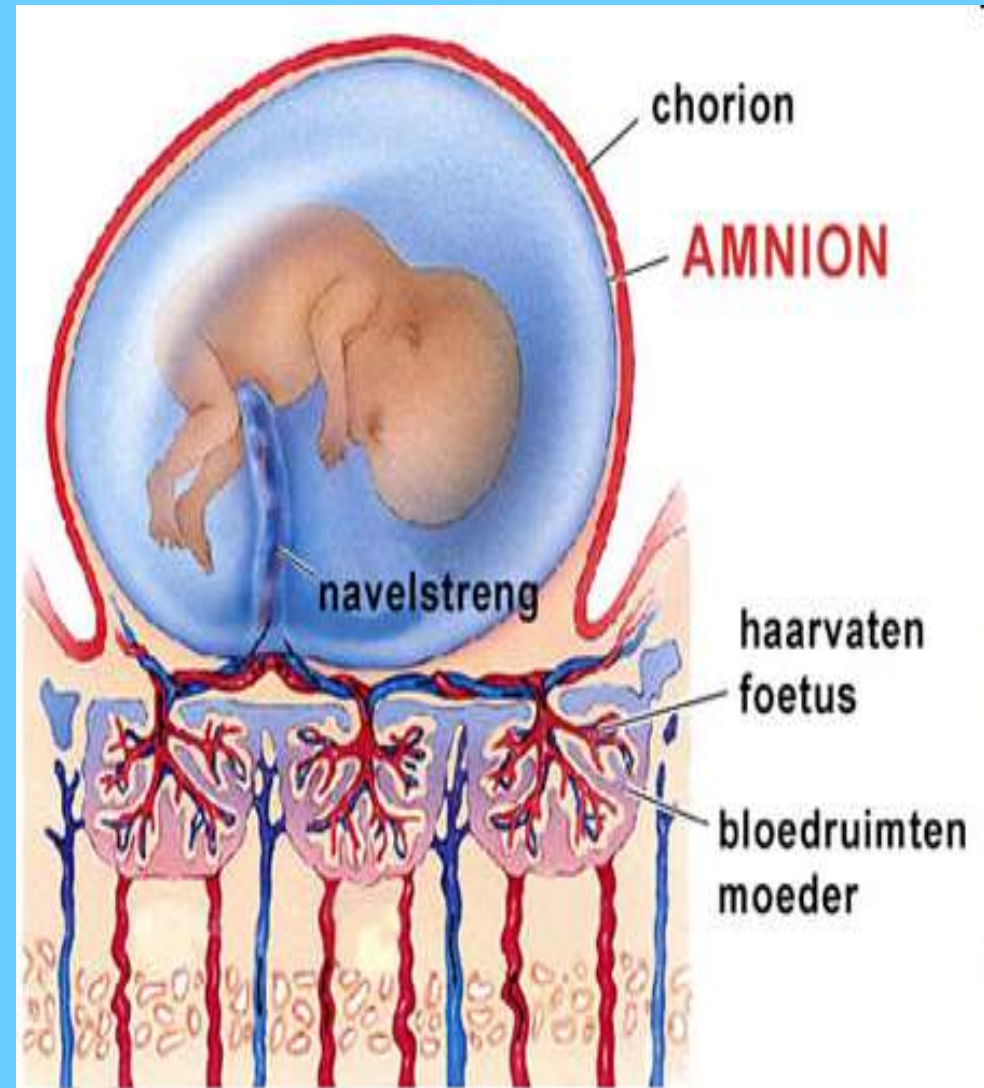


- **Chorionic villi are the functional units of placenta.**

Amnion



- Derived from **ectoderm**.
- Completely **covers the embryo and lines the fetal aspect** of placenta.
- With embryo growth the amnion obliterates the chorionic cavity.
- **Amnionic sac** is fluid filled called amnionic fluid: the embryo is bathed in the fluid.



Amniotic fluid



- Plays a major role in **fetal growth and development.**
- The amniotic fluid is derived from the **amniochorionic membrane**, through the **fetal skin**, from the **respiratory tract and urinary tract.**
- The fetus swallows about **400ml of amniotic fluid a day in the final stages** of pregnancy.



Amniotic fluid is the clear, yellowish fluid that surrounds and protects the fetus in the uterus

Significance of Amniotic Fluid



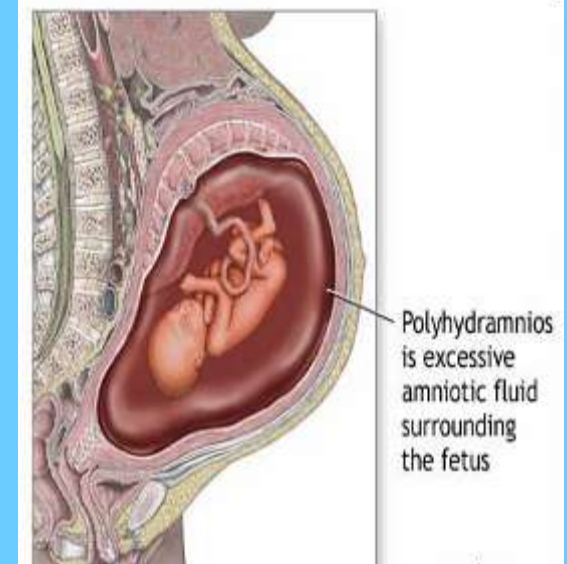
- Permits **symmetric external growth** of the embryo and fetus.
- Acts as a **barrier to infection**.
- Permits **normal fetal lung development**.
- Helps **control the embryo's body temperature** by maintaining a relatively constant temperature.
- **Cushions** the embryo and fetus against **injuries** by distributing impacts the mother receives.



Birth Injuries Relating To Amniotic Fluid



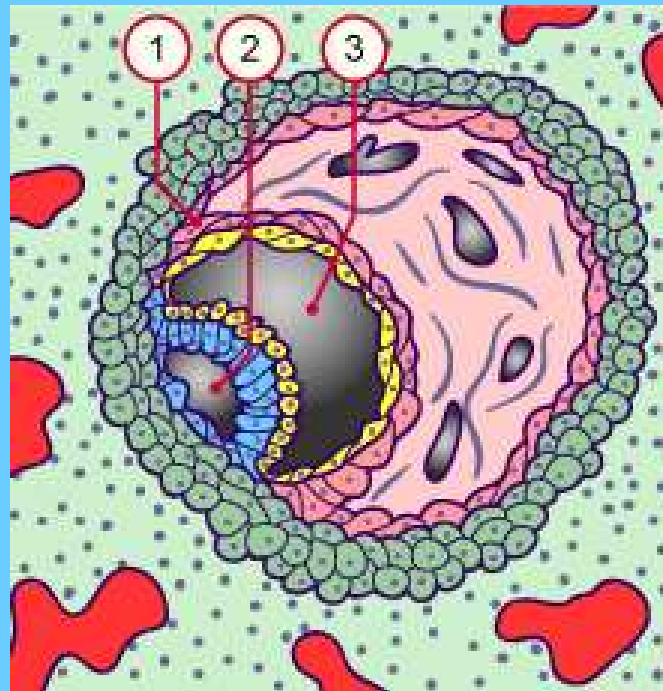
- An abnormally High levels of amniotic fluid, is called polyhydramnios (**>2000ml**).
- Low levels of amniotic fluid (**<1000ml at term**) is called oligohydramnios.
- Both conditions are associated with congenital anomalies (birth defects).



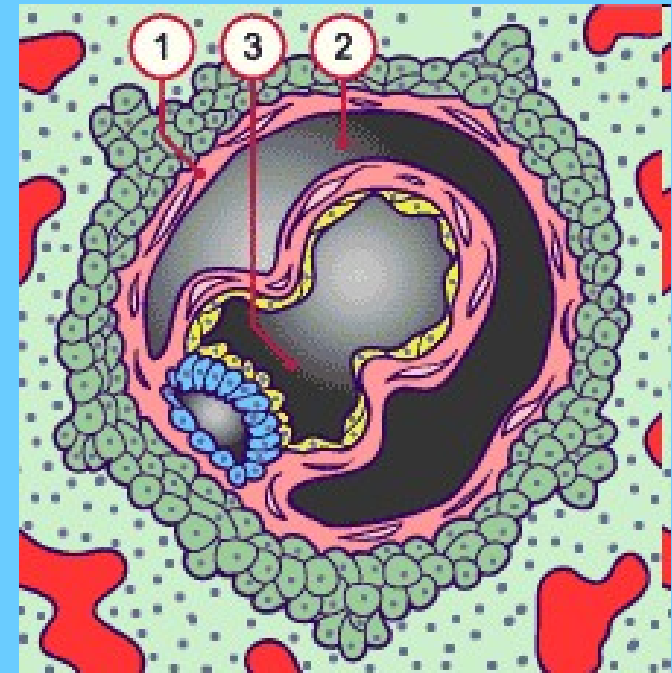
The Umbilical Vesicle (Yolk Sac)



- A **membrane** outside the embryo that is connected by a tube (the yolk stalk) through the umbilical opening to the embryo's midgut.



- 1.Extraembryonic mesoblast
- 2.Amniotic cavity
- 3.Primary yolk sac
(= primary umbilical vesicle).

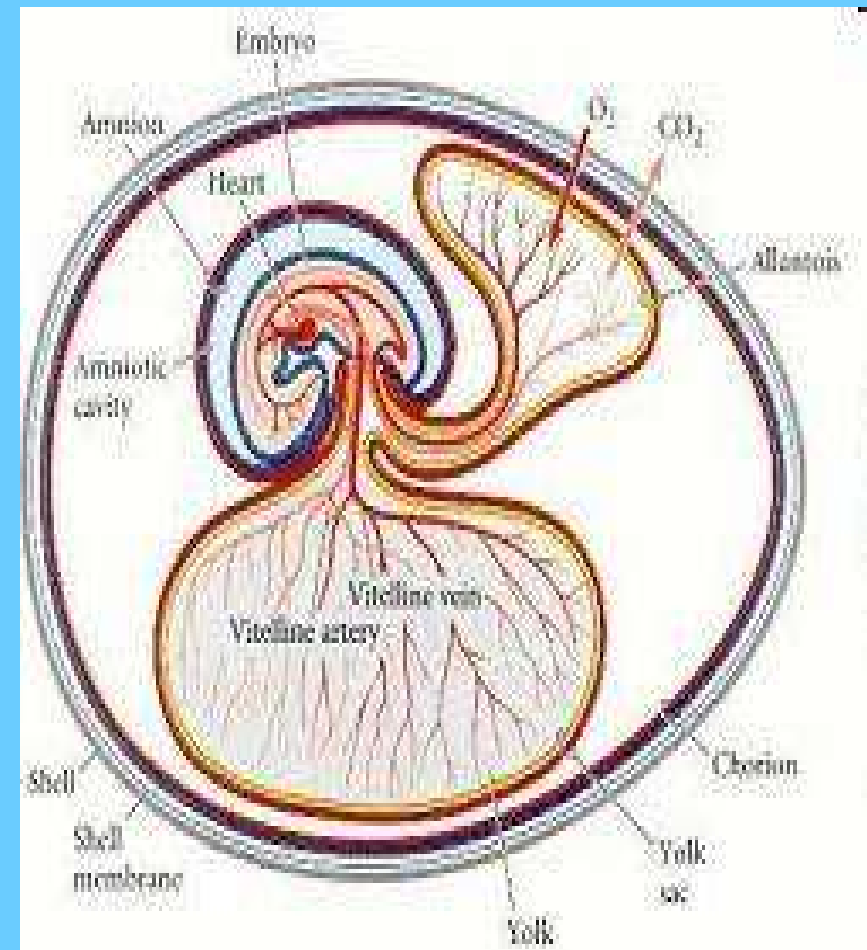


- 1.Extraembryonic mesoblast
- 2.Chorion cavity
- 3.Secondary yolk sac
(= secondary umbilical vesicle)

Significance



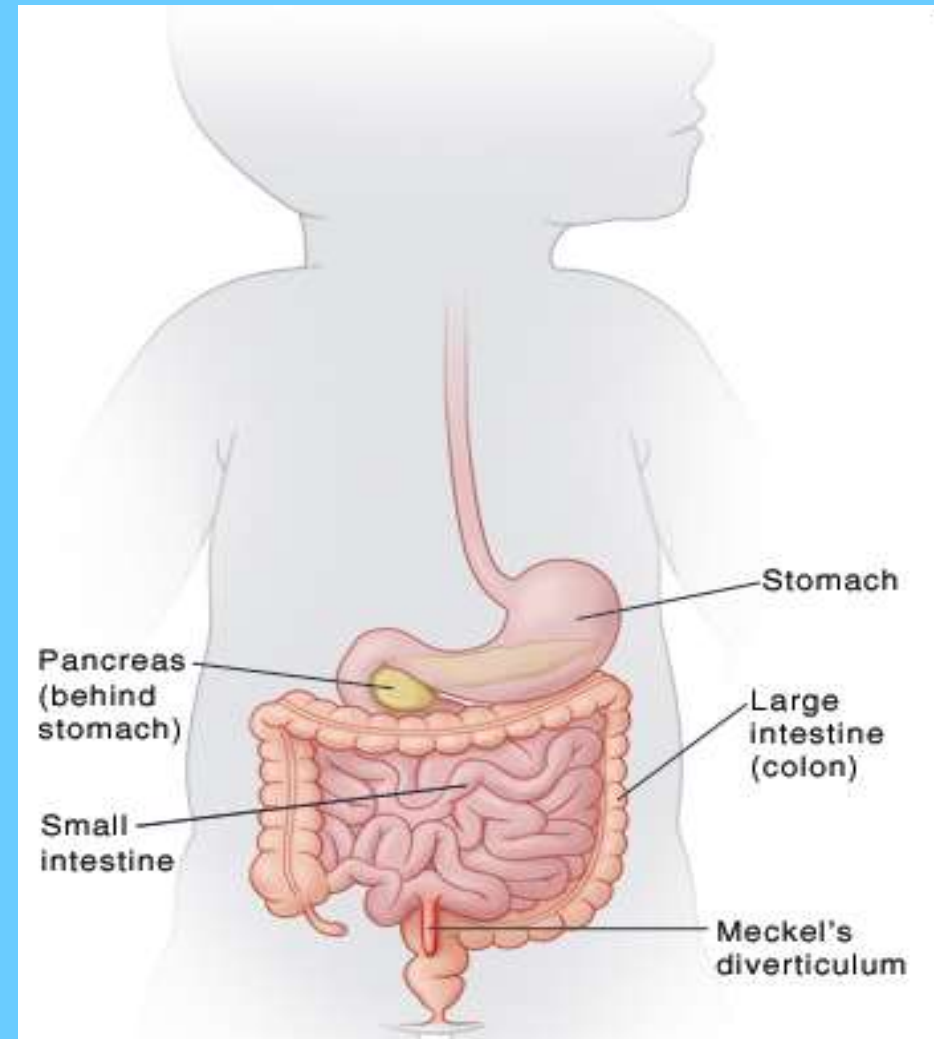
- **Transfer of Nutrients to the embryo** (2nd and 3rd weeks).
- **Primitive Gut** (During 4th week).
- **Blood development** (mesoderm - 3rd to 5th week).
- **Primordial germ cells** (endoderm , 3rd week)





Fate of the Umbilical Vesicle

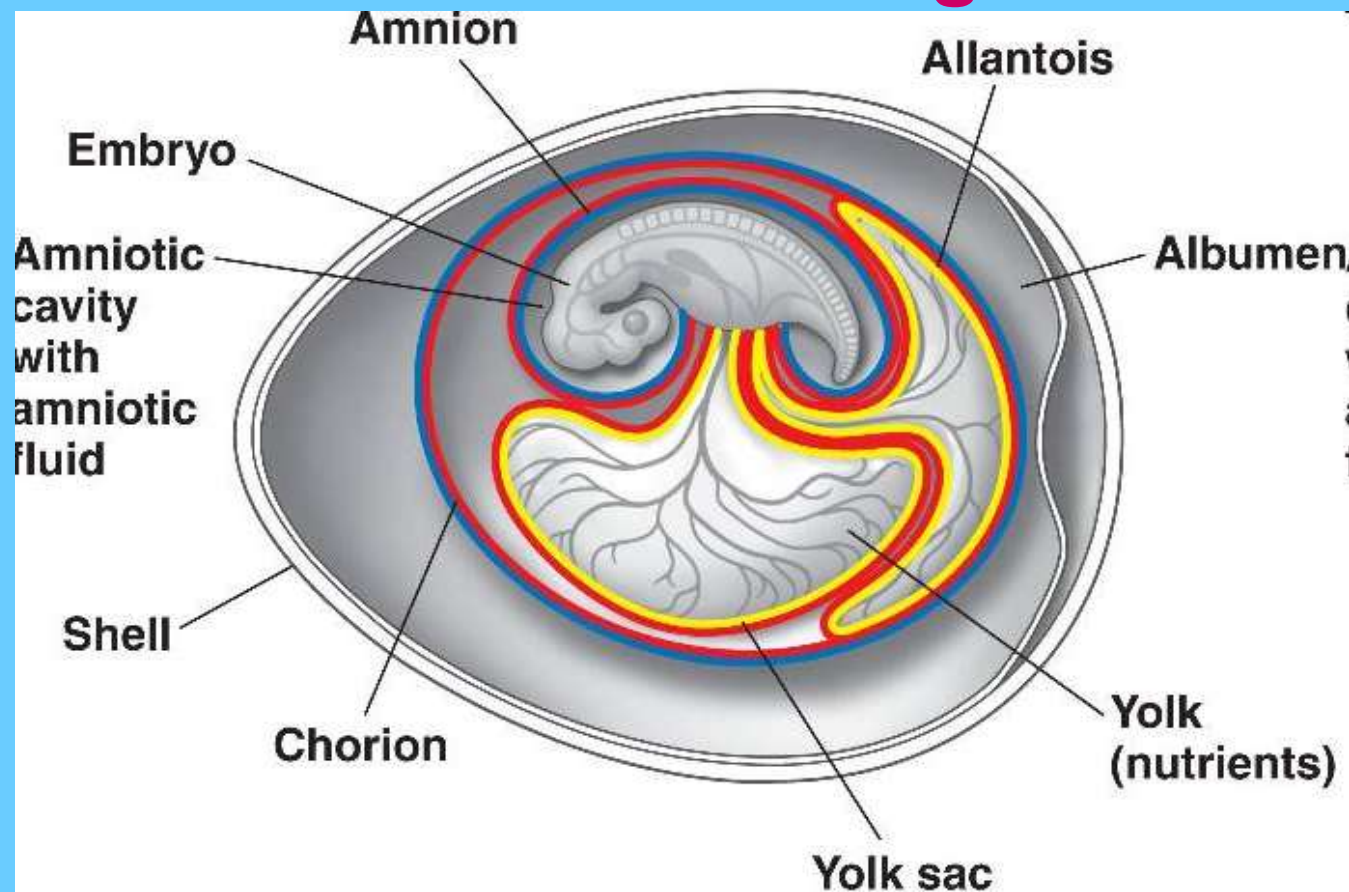
- **Atrophies** as pregnancy advances.
- later **shrinking** -
- **non-functional** – **Meckel's diverticulum** (outpocketing of small intestine)





Allantois

- Sausage like diverticulum from the caudal wall of the umbilical vesicle that extends into the connecting stalk.



Significance

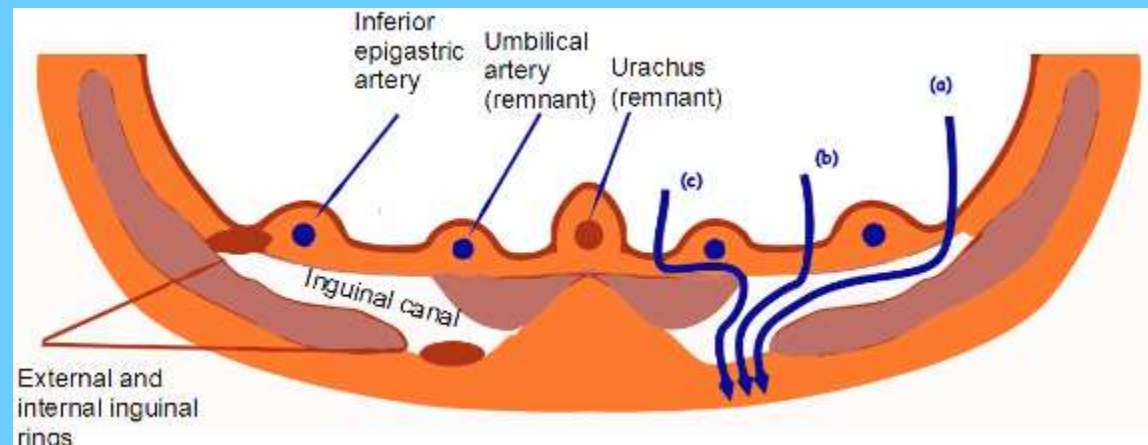


- **Blood formation** (3rd to 5th weeks).
- **Blood vessels persist** as umbilical vein and arteries.
- Part of **urinary bladder**.



Fate:

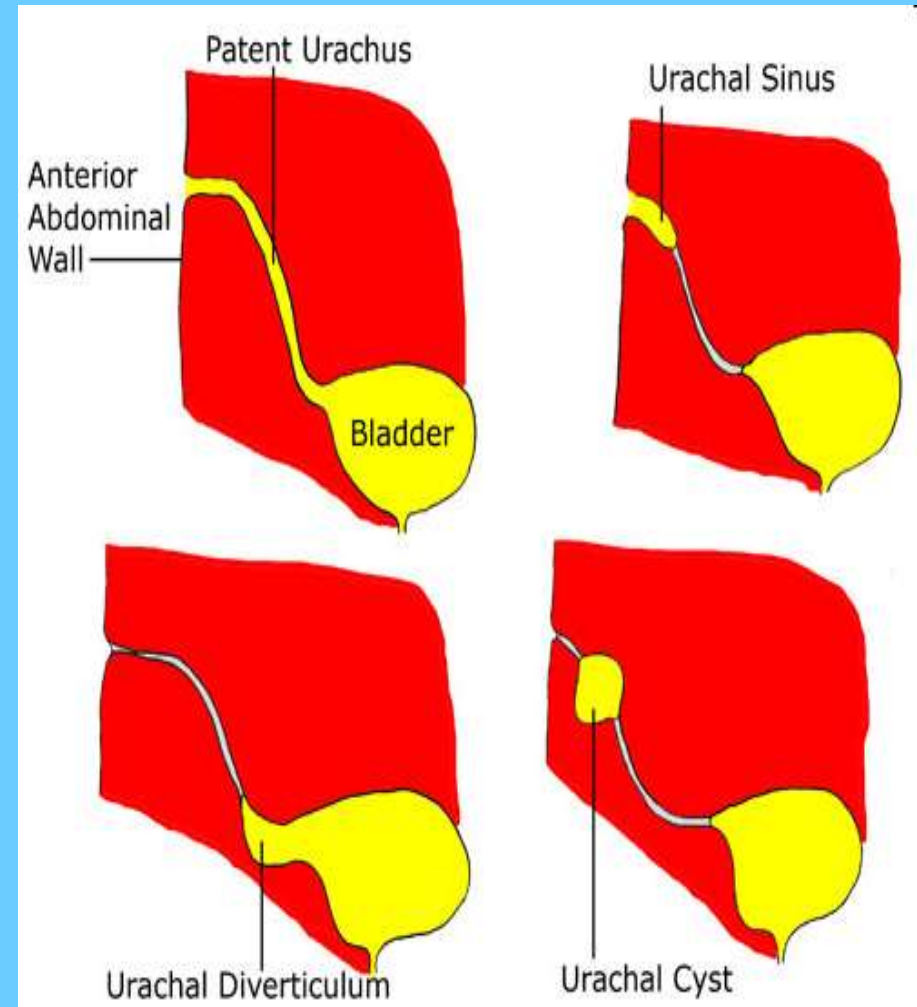
- **Urachus** – median umbilical ligament.





Urachus and anomalies

- A partially obliterated allantois in fetal life is called **Urachus**.
- This will obliterate postnatal and becomes the **median umbilical ligament**.
- Any failure to do so results in **patent Urachus**(weeping umbilicus)

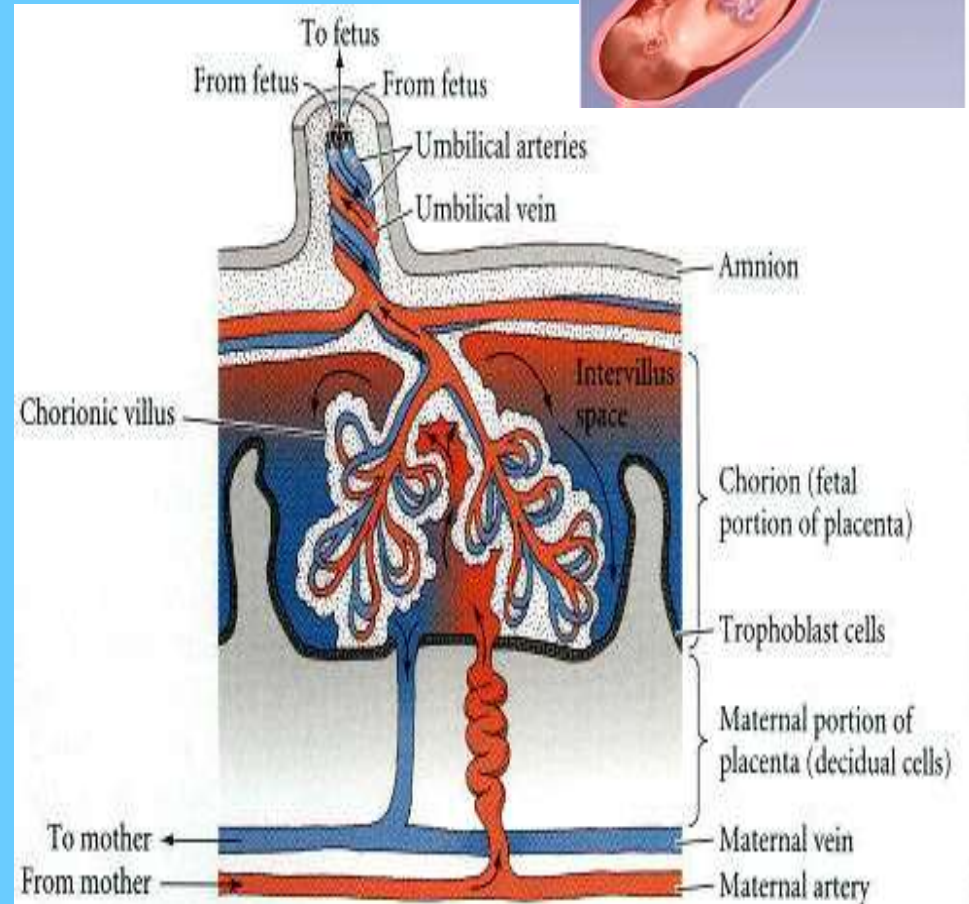
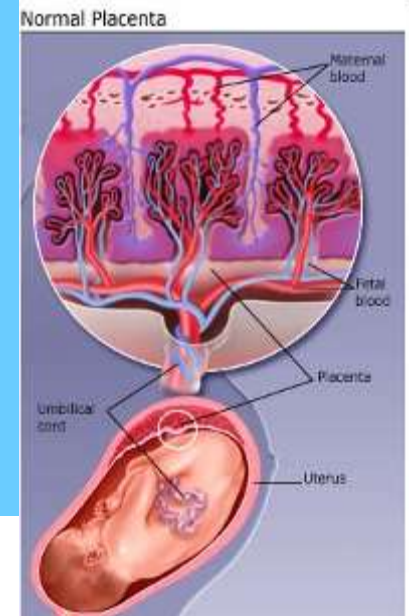


The Placenta

- Primary site of nutrient and gas exchange.

Components:

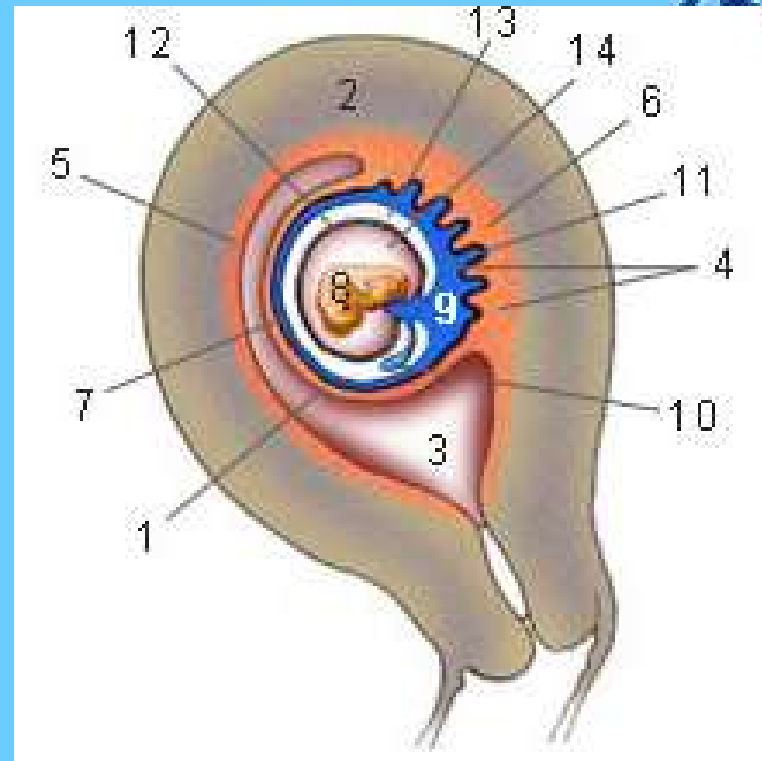
- A **fetomaternal organ** having two components:
 - Fetal component : **the villous chorion.**
 - Maternal Component: **the decidua basalis.**



The Decidua



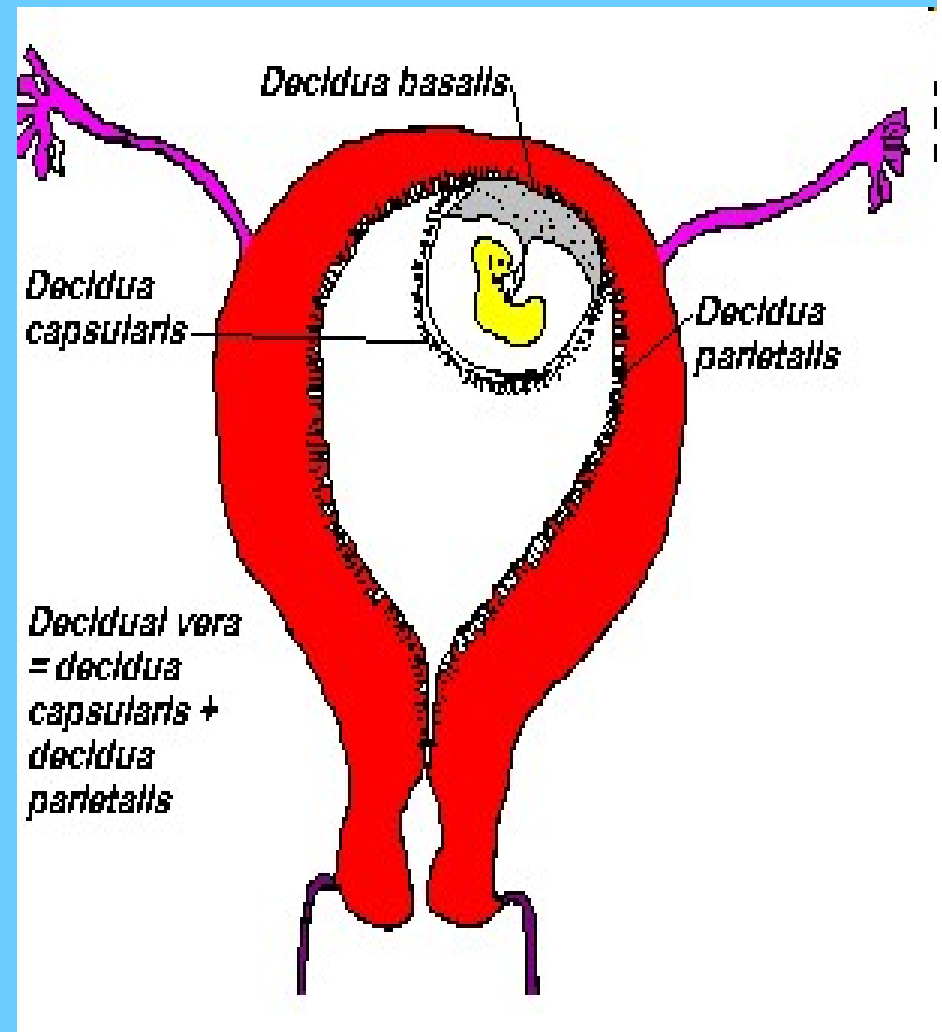
- **Decidua refers to gravid endometrium, the functional layer of the endometrium in a pregnant women.**
- **Separates from the uterus after childbirth.**



1. Decidua capsularis
2. Uterine wall
3. Uterine cavity
4. Placenta
5. Decidua parietalis
6. Decidua basalis

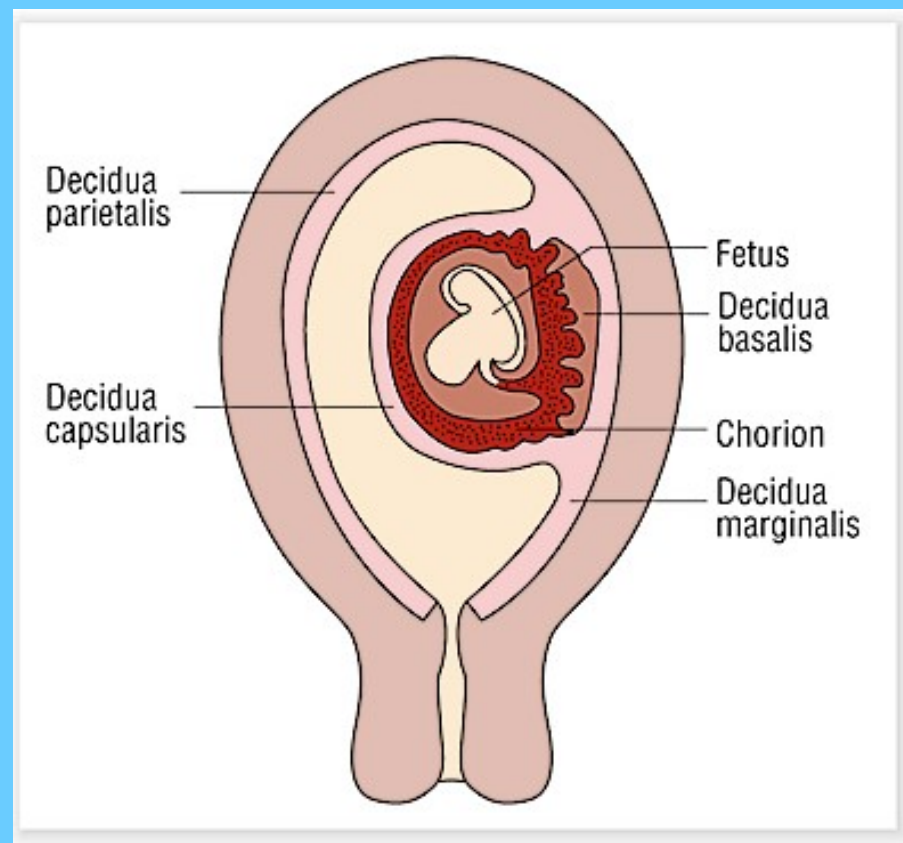
Three regions of decidua

- **Decidua Basalis:** Part of decidua deep to the conceptus.
- **Decidua Capsularis:** Superficial part of the decidua overlying the conceptus.
- **Decidua Parietalis:** all remaining parts.

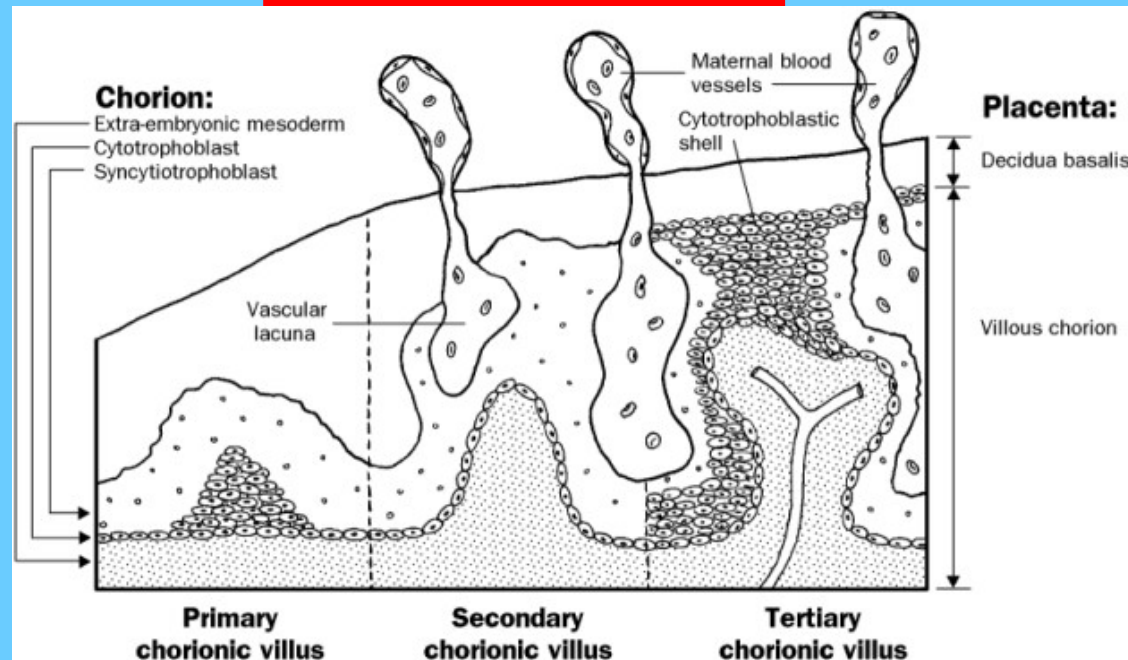


Shape

- Discoid, 15 to 20 cm in the diameter, 2 to 3 cm thickness, 500 to 600 gm weight.



Structure



- **Primary chorionic villi** are solid outgrowths of cytotrophoblast that protrude into the syncytiotrophoblast.
- **Secondary chorionic villi** have a core of loose connective tissue, which grows into the primary villi about the third week of development.

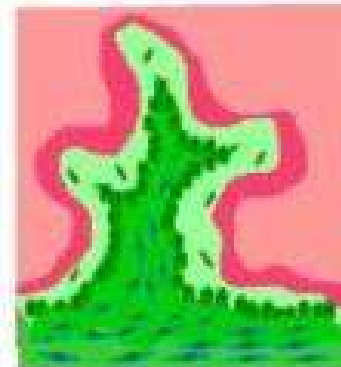
Structure



- Tertiary chorionic villi contain embryonic blood vessels that develop from mesenchymal cells in the loose connective tissue core.
- These blood vessels connect up with vessels that develop in the chorion and connecting stalk and begin to circulate embryonic blood about the **third week** of development.



Primary villi consist of syncytiotrophoblast and cytotrophoblast day 13



Secondary villi have a core of extra-embryonic mesoderm day 16

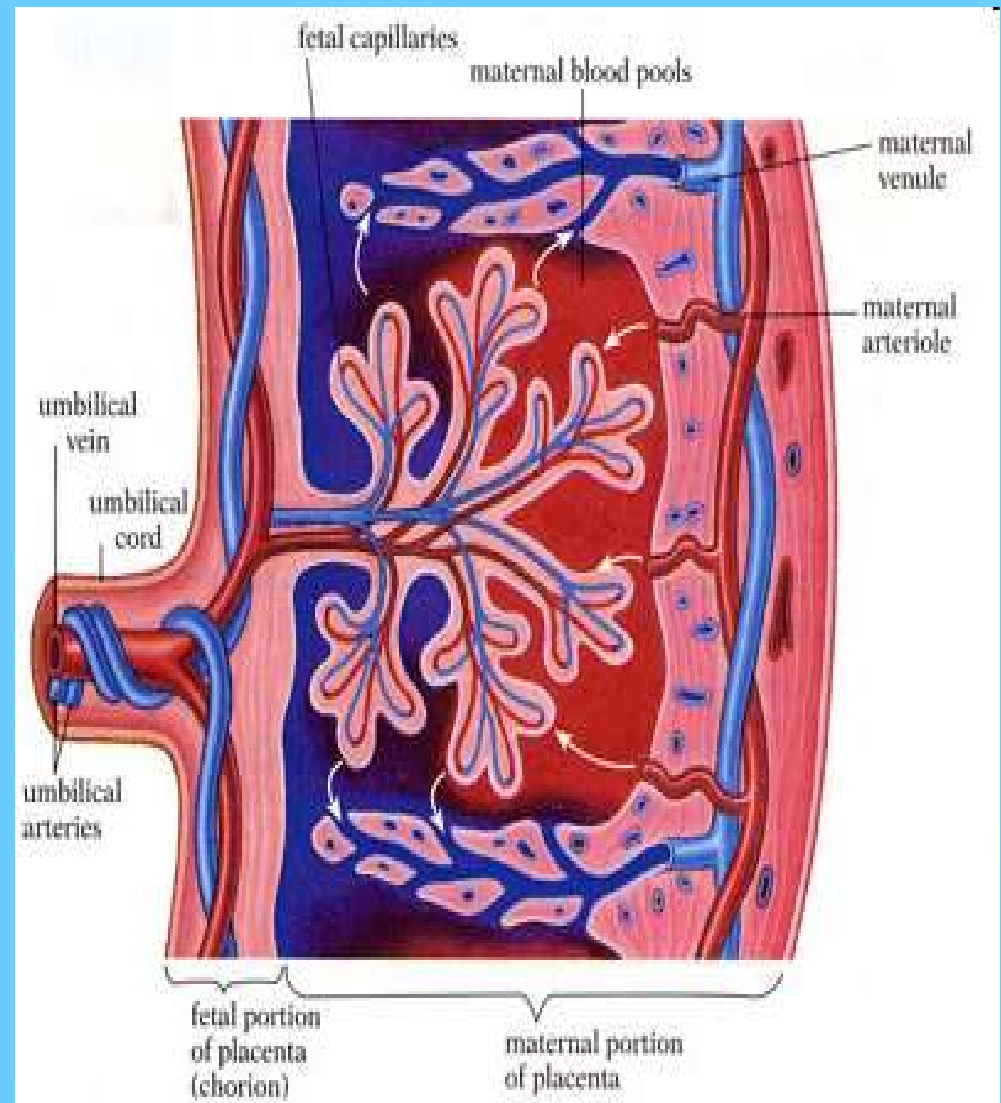


Tertiary villi have blood vessels in the mesoderm day 21

The fetal part of placenta

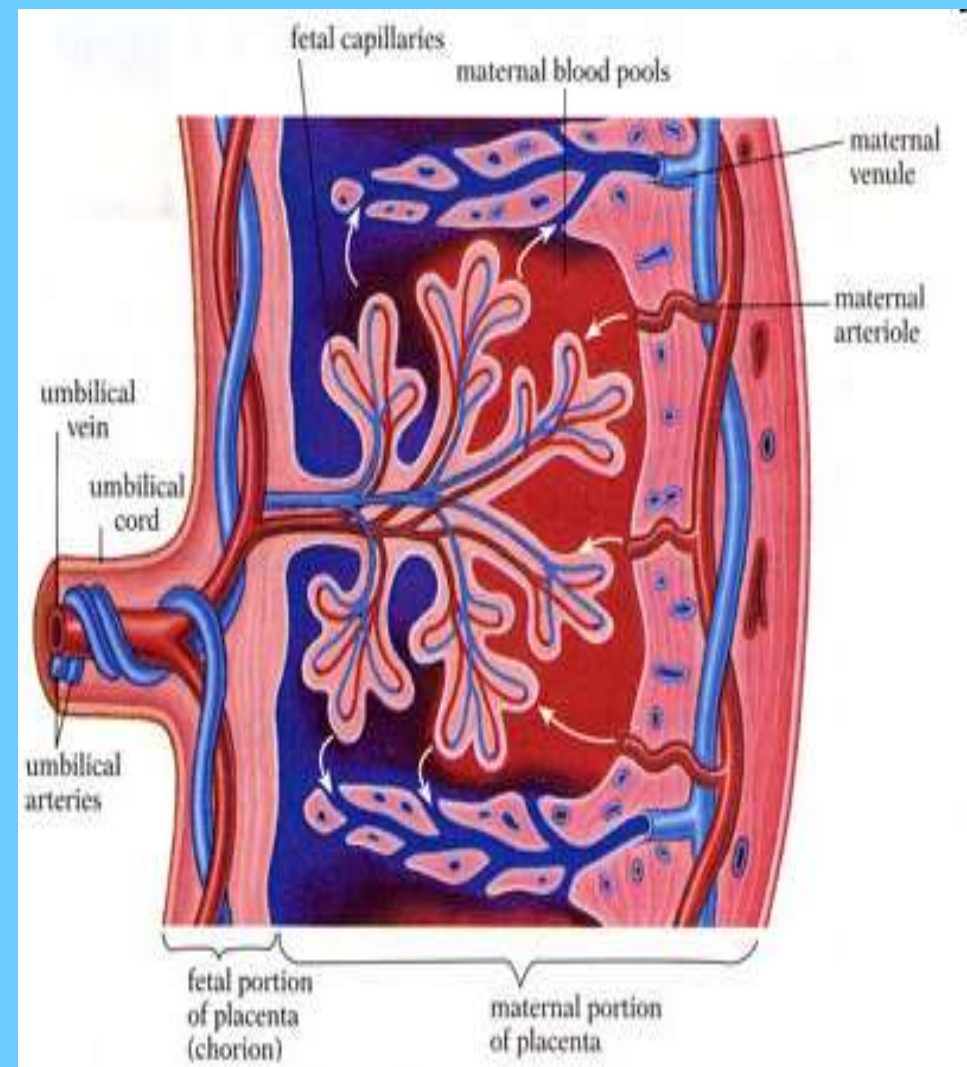


- fetal portion of the placenta consists of the villi of the **chorion frondosum**;
- These villi are suspended in the intervillous space, and bathed in maternal blood.
- It is the **chorionic portion of the placenta**, containing the fetal blood vessels, from which the umbilical cord arises.



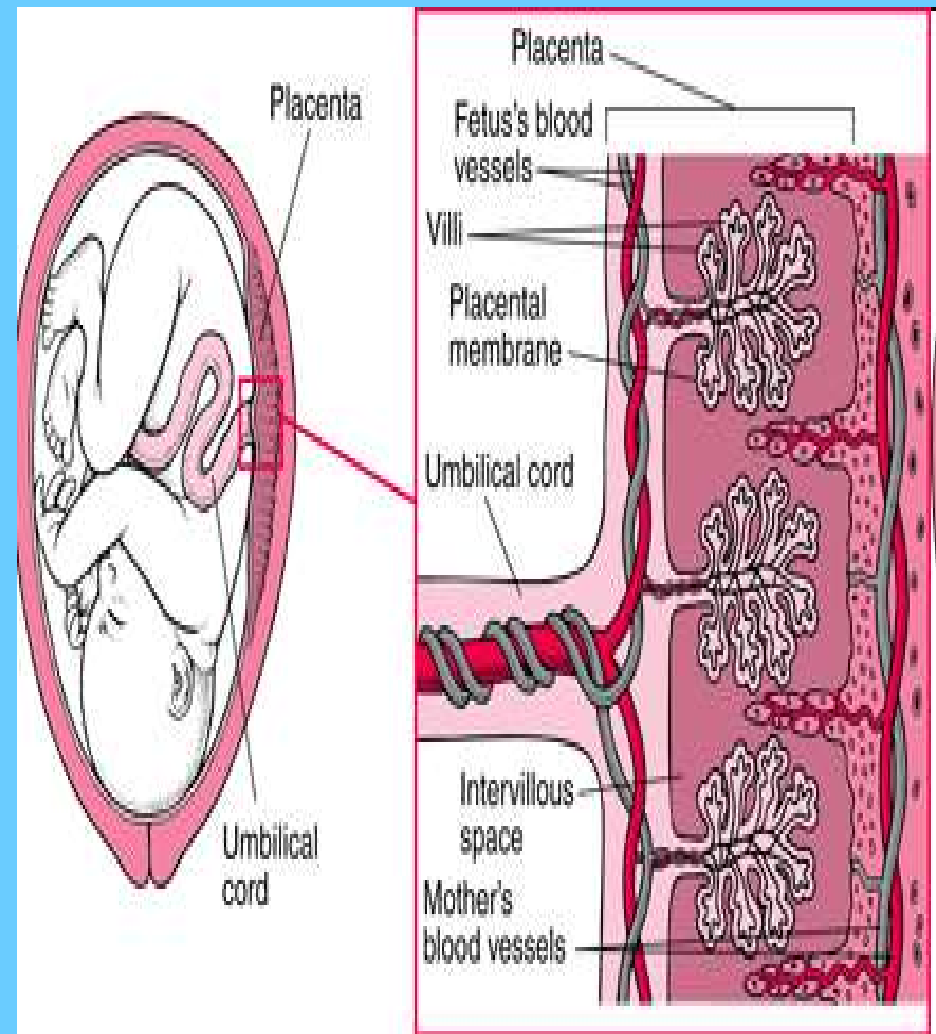
Maternal Portion.

- Formed by the decidua basalis containing the **intervillous space.**
- From the maternal side the placenta has a wrinkled surface, characterized by the **cotyledons.**



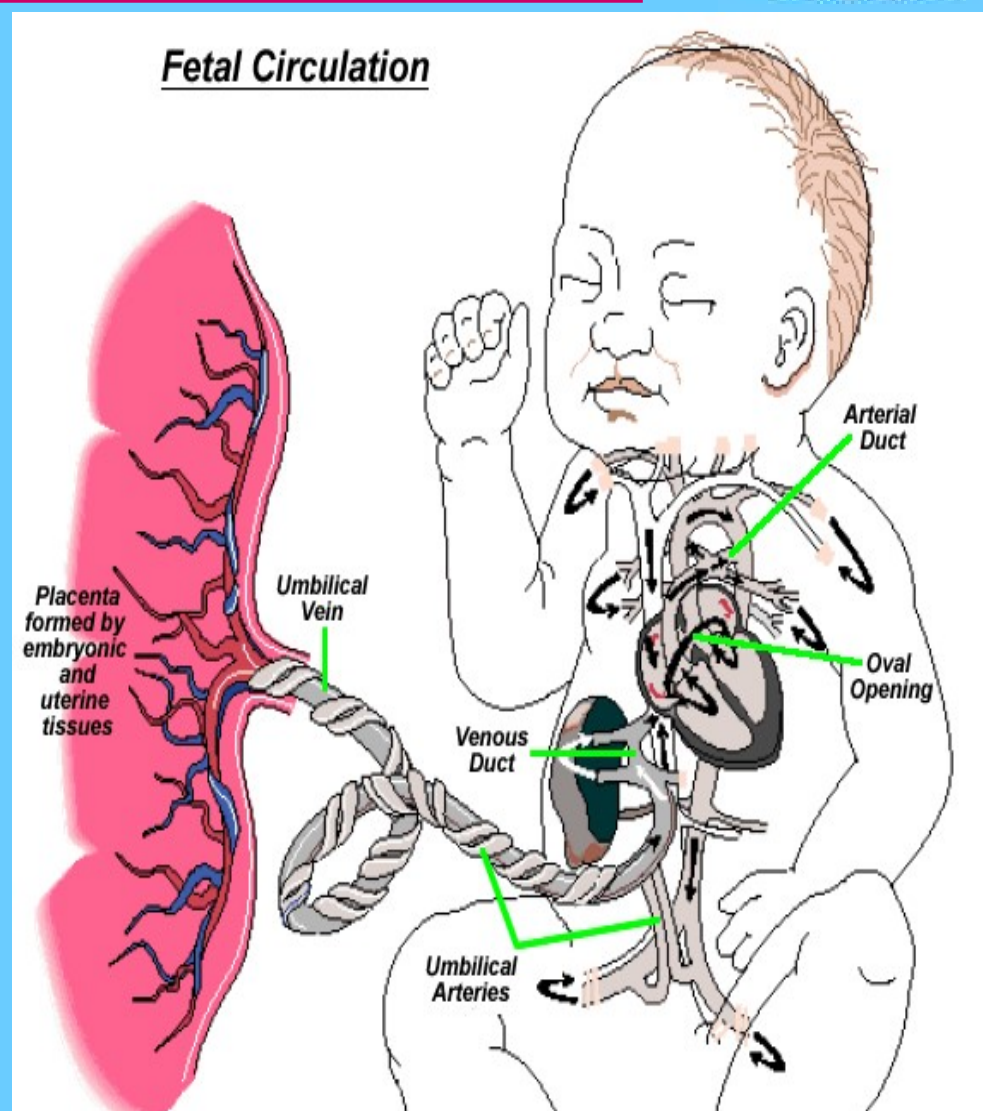
Intervillous Space

- **Intervillous space** of the placenta contain maternal blood.
- Derived from the **lacunae** that developed in the syncytiotrophoblast during the second week.
- This large blood – filled space results from the coalescence and enlargement of the lacunar networks.



Placental Circulation

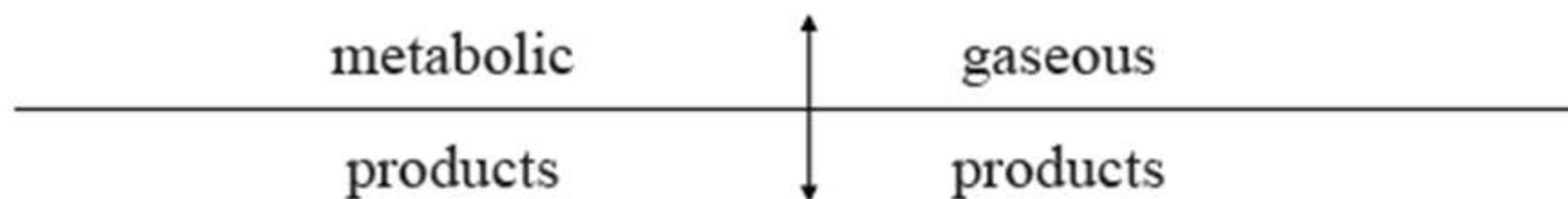
- Circulation of blood through the placenta during intrauterine life, serving fetal needs for aeration, absorption, and excretion.
- Placental blood flow is increased at term and amounts to **500 ml/min (80% of the uterine perfusion)**.



Placental circulation

fetal placental circulation
(within the villus)

umbilical A. → arterio-capillary-venous system → umbilical V.



spiral A. → intervillous space → endometrial V.

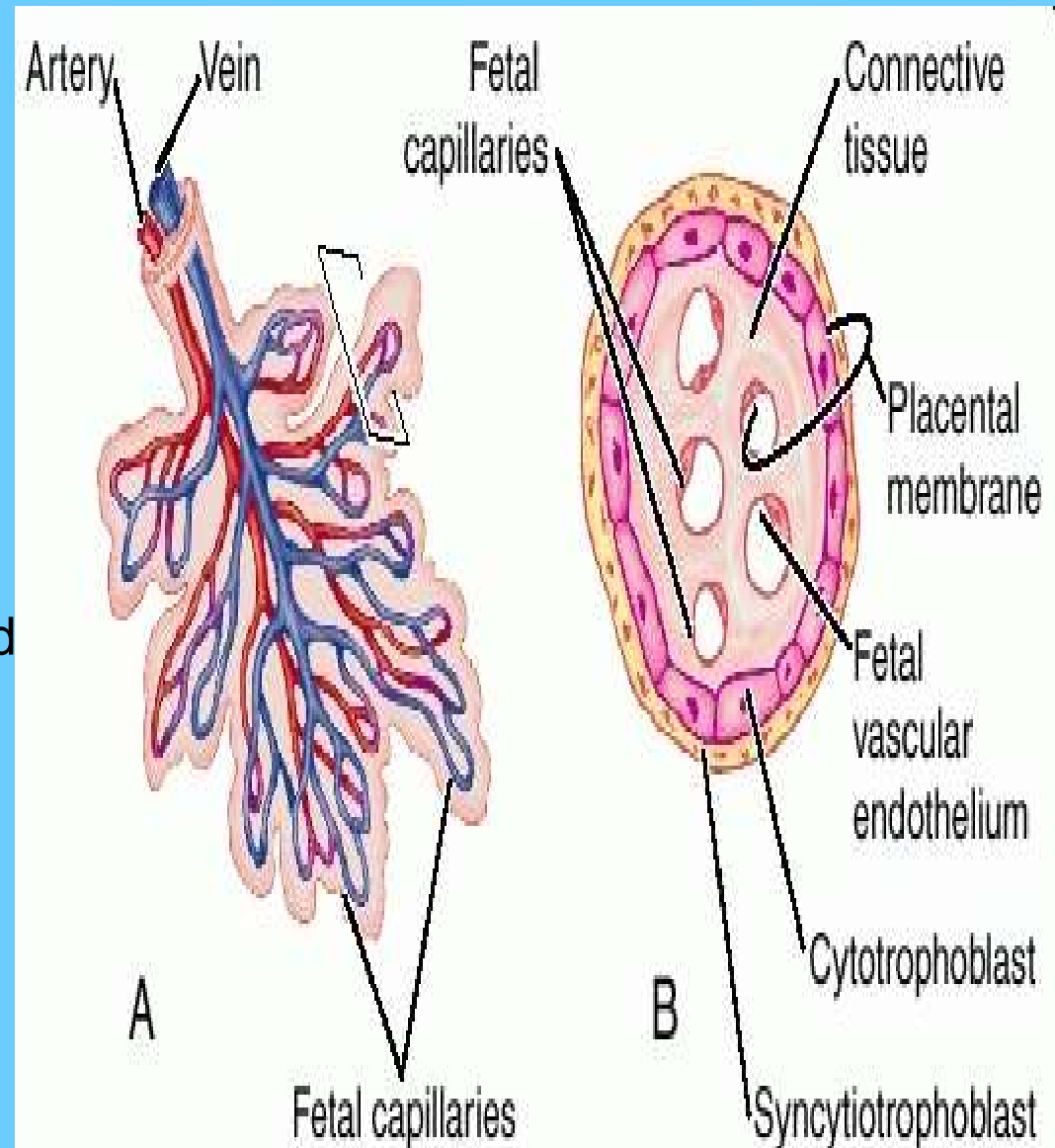
maternal placental circulation

intrauterine growth retardation (IUGR)

The placental barrier



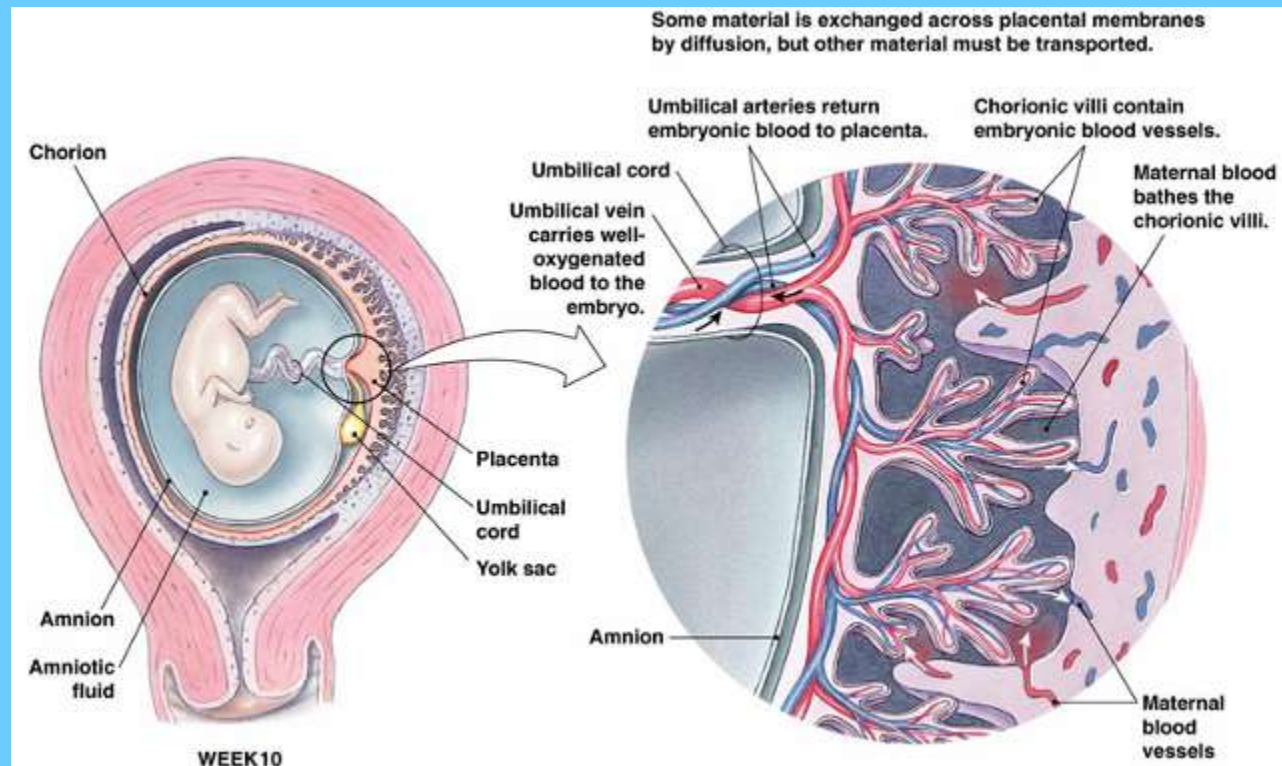
- The placental barrier is composed of structures that separate the maternal and the fetal blood.
- **In the first trimester** it consists of:
 - the syncytiotrophoblast,
 - the cytotrophoblast (Langhans' cells),
 - the villus mesenchyma and
 - the fetal capillary walls.
- **During the 4th month** the cytotrophoblast disappears from the villus wall



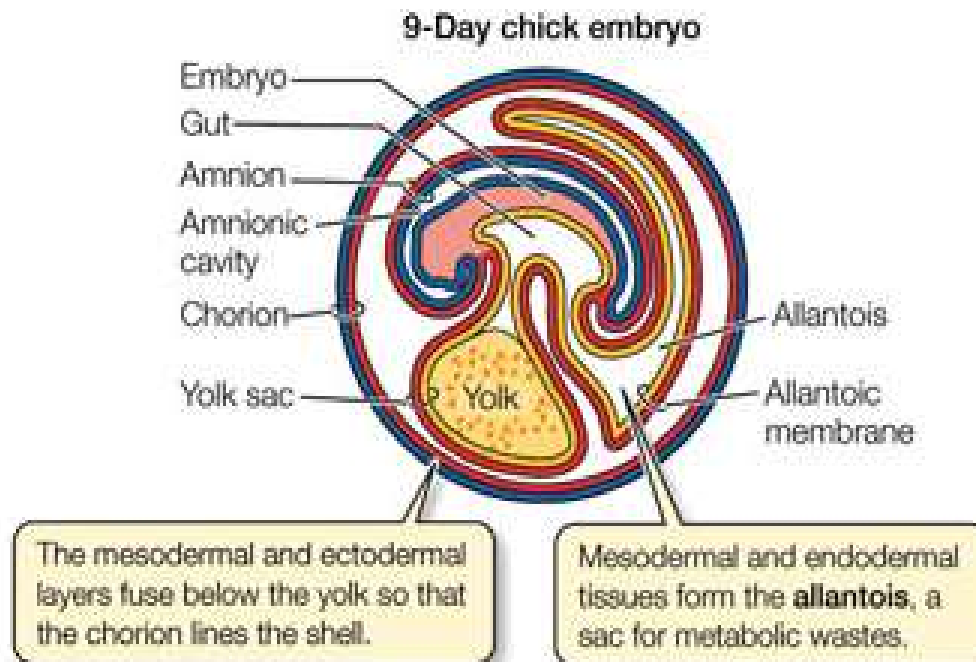
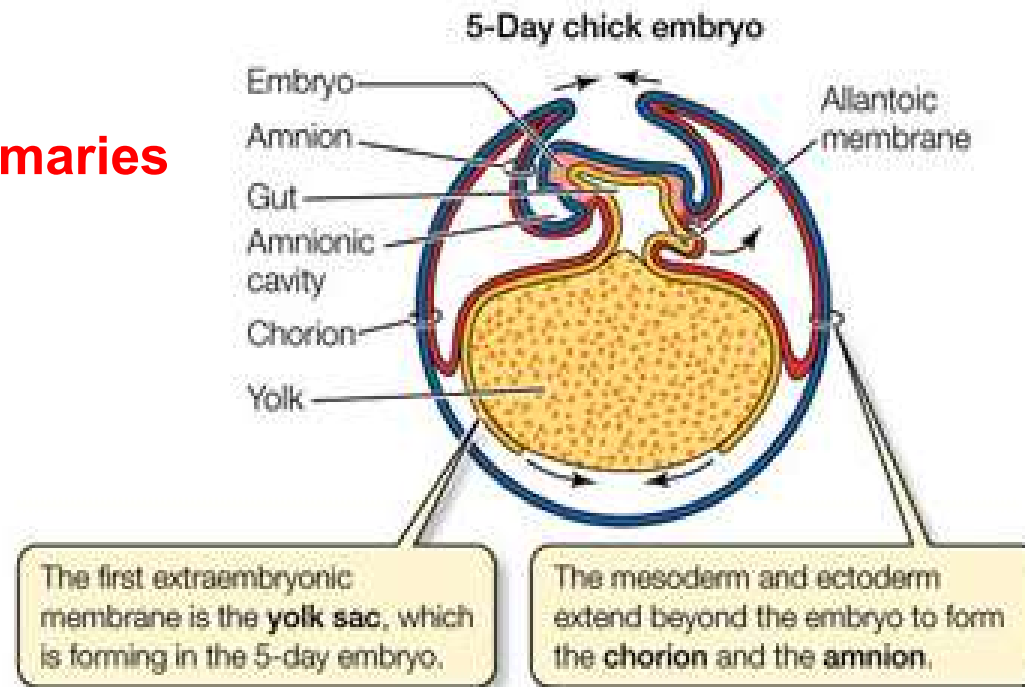
Functions of the placenta



- Metabolism.(synthesis of glycogen).
- Transport of gases and nutrients.
- Endocrine secretion (human chorionic gonadotrophin - hCG.)



Interactive Summaries





References

- The Developing Human
Clinically Oriented Embryology.
Eighth Edition. Keith .L. Moore.
T.V.N.Persaud.